Recording

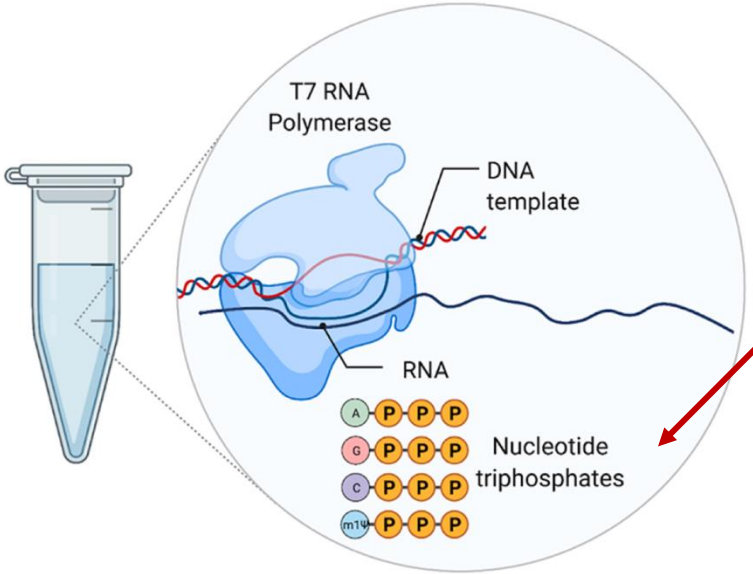
EN 580.133
siRNA as Therapeutics (Part I)

Elaine Zhelan Chen

Jan 9, 2025

Created by Elaine Zhelan Chen

Problem Set #1
In vitro transcription (IVT)



Make sure to include the modified nucleotides, i.e., pseudouridine

T7 RNA polymerase, DNA template, and <u>nucleotides</u>

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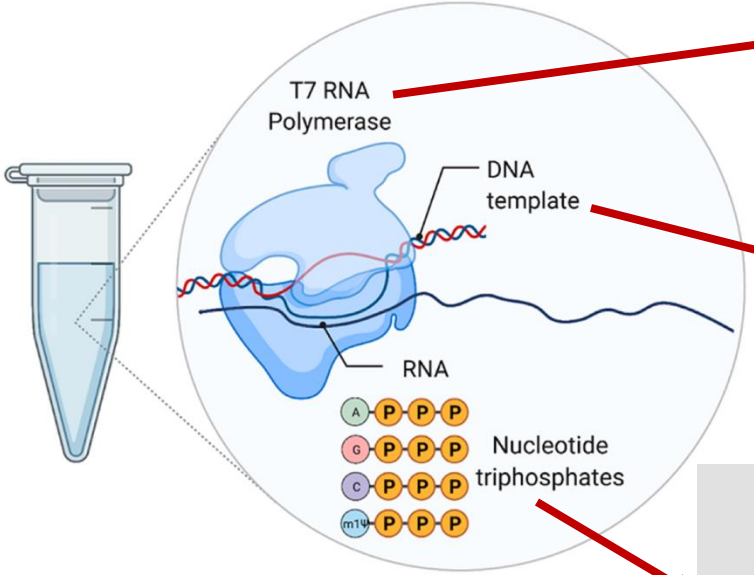
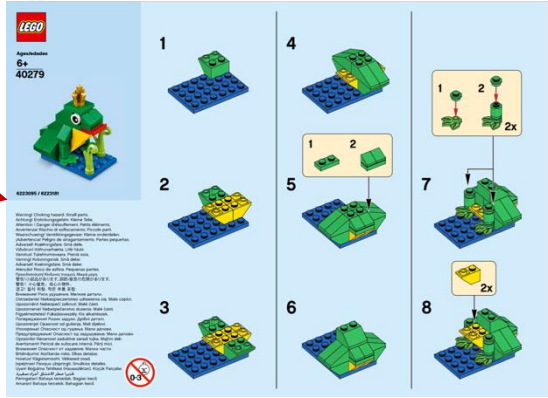




Recap

- Applications of mRNA therapeutics
  - Immunogenic
    - Prophylactic vaccines
    - Individualized neoantigen therapies (INTs)
  - Non-immunogenic
    - Protein replacement therapies
- Mechanism of mRNA vaccine, roles of somatic cells, APCs, B cells, CD8+ T cells, and CD4+ T cells
- Designs of 5’UTR, 3’UTR, and poly(A) tail in the COVID vaccines
- The process of creating a personalized cancer vaccine and the role of next generation sequencing
- mRNA protein replacement therapy and its unique set of challenges

Emergence of siRNA therapeutics

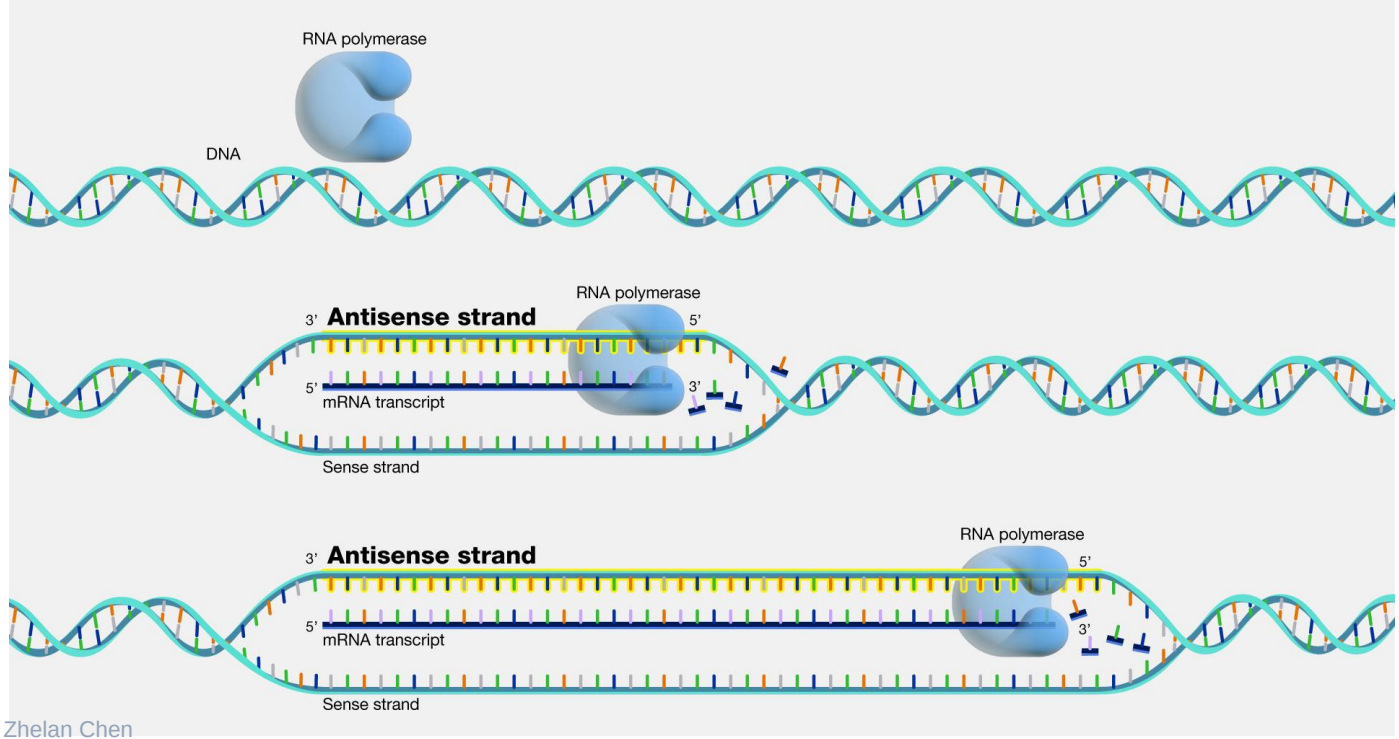
siRNA

- RNA interference (RNAi) was discovered in 1998
- First use of RNA interference to inhibit hepatitis C virus replication in mice in 2002
- First siRNA drug clinical trial in 2010
- First siRNA drug approved in 2018
    - ONPATTRO (Patisiran) for treating Hereditary Transthyretin Amyloidosis (hATTR)
- A total of 6 siRNA drugs were approved from 2018-2024

siRNA

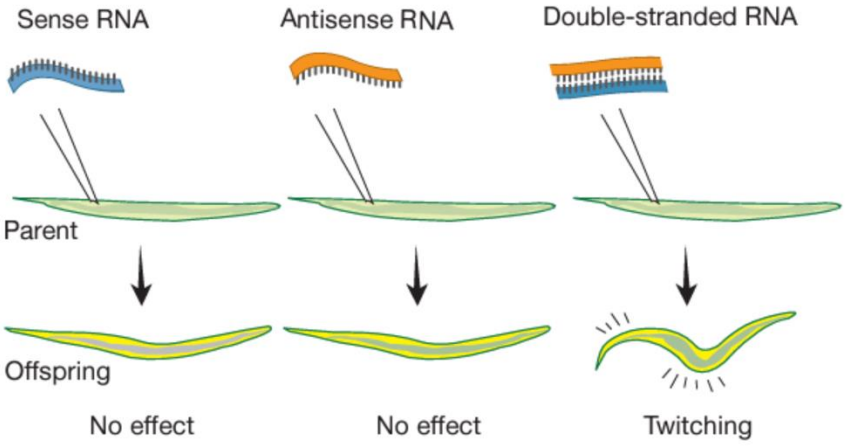
Sense and Antisense strand

- The genetic code in mRNA is the sense sequence
- The antisense RNA can base-pair with the mRNA



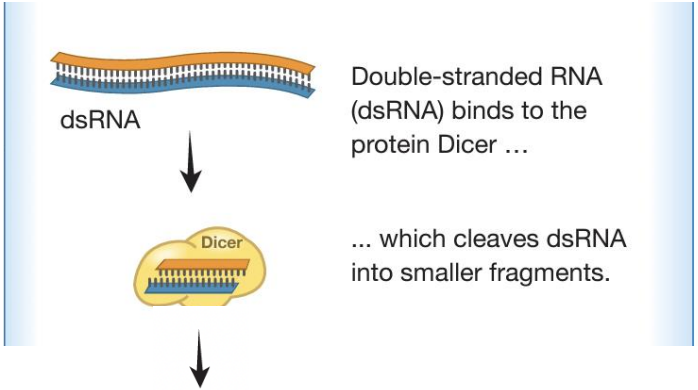
Discovery of RNA interference (RNAi)

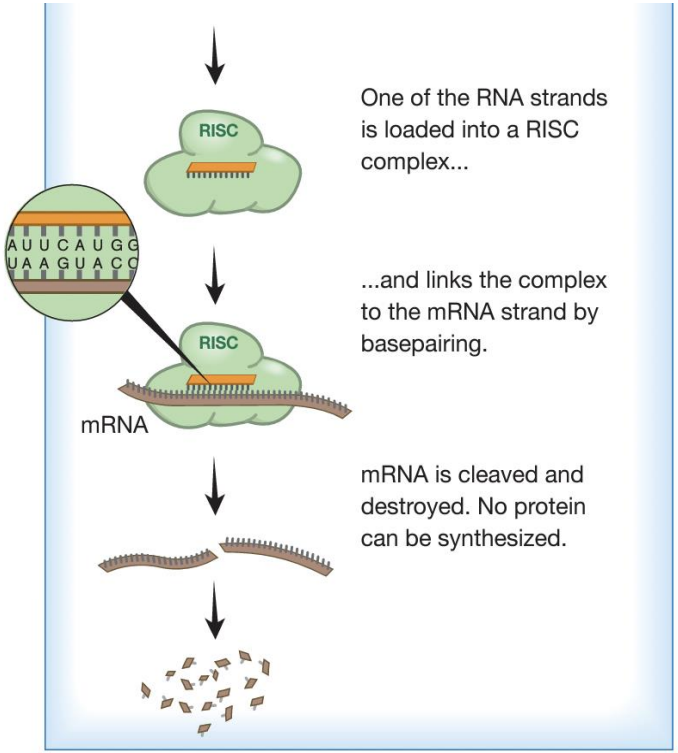
- Fire and Mello investigated how gene expression is regulated in the nematode worm (C. elegans)
- They injected sense RNA, mRNA encoding a muscle protein, which led to no changes in the behavior of the worms
- The injected antisense RNA, RNA that can base pair with the mRNA, which also led to no change
- But when they injected sense and antisense RNA together, dsRNA, they observed that the worms displayed twitching movements
- Similar movements were seen in worms that completely lacked a functioning gene for the muscle protein



RNA interference (RNAi)

- RNA interference (RNAi) is a biological mechanism in regulation of gene expression

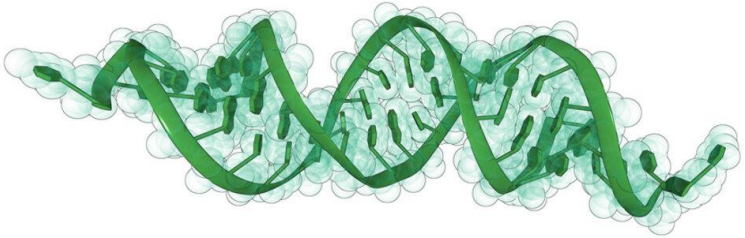


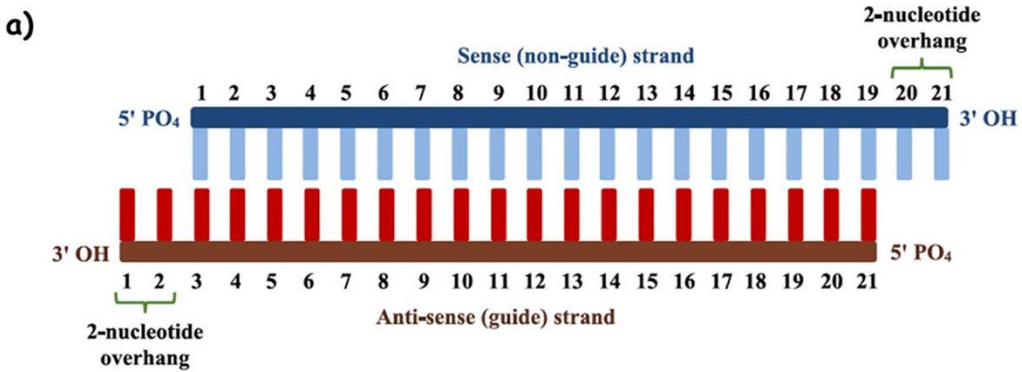


Introduce tailor-made dsRNA and use the RNAi machinery to degrade specific mRNA by sequence complementarity

siRNA structure

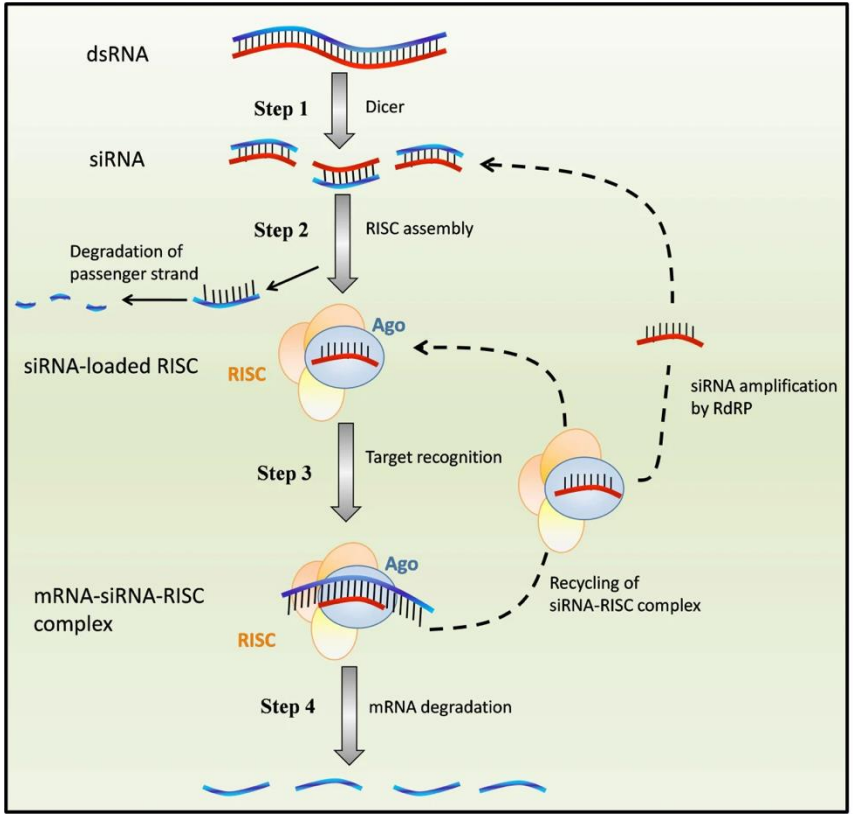
- Small interfering RNA
- Double-stranded oligonucleotides
  - Guide (antisense) strand
  - Passenger (sense) strand
- Typically, around 21-23 base pairs long
  - Phosphorylated 5' ends
  - Hydroxylated 3' ends
  - Two overhanging nucleotides at the 3' end of each strand





siRNA-mediated gene silencing

- siRNAs degrade target mRNA via sequence complementarity to halt subsequent protein production
- Upon entering cells, siRNA is loaded into the argonaute 2 (Ago2), which recruits other proteins to form an active RNA-induced silencing complex (RISC)
- The passenger strand is then released
- The guide strand-loaded Ago2 identifies and degrades the target mRNA

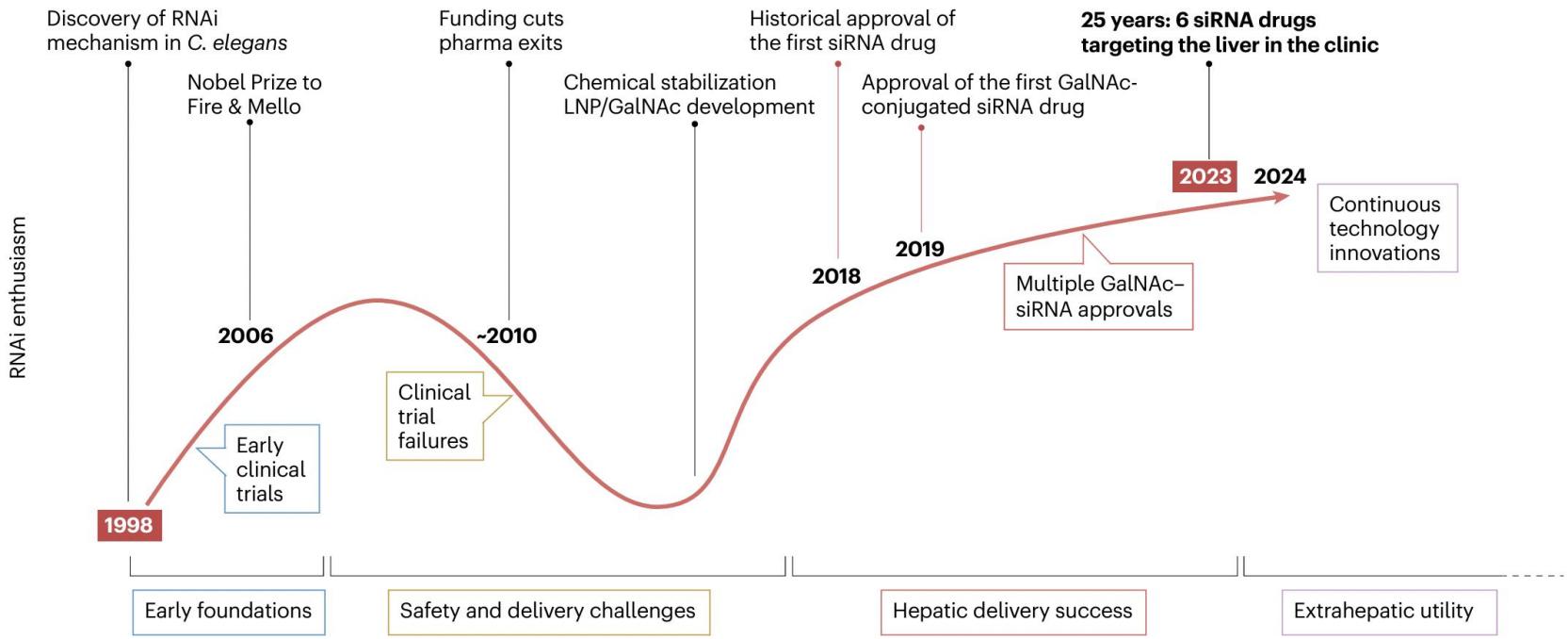


siRNA therapeutics

Advantages of siRNA as drugs

- siRNA drugs can target previously undruggable targets at the RNA level through a simple base-pairing mechanism
- siRNA drug pipeline costs significantly lower than conventional drug pipelines
- siRNA drug development timeline is shorter
- siRNA has shown multi-month durability of therapeutic effects

siRNA drug development



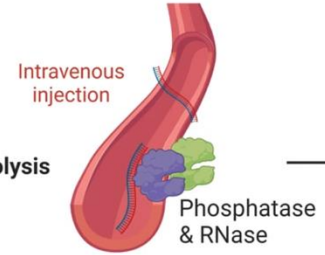
Created by Elaine Zhelan Chen

Challenges in siRNA design

Mismatch
leads to off-target effect

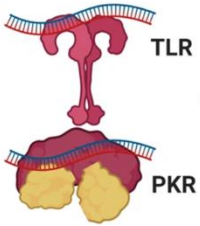
→ **Translational supression**
Off-target mRNA decay

Instability
to degradation & hydrolysis



→ **Enzymatic &**
Non-enzymacic hydrolysis

Innate immunity
recognized by TLR3 & PKR



→ **Global translational**
blockade & Cell death

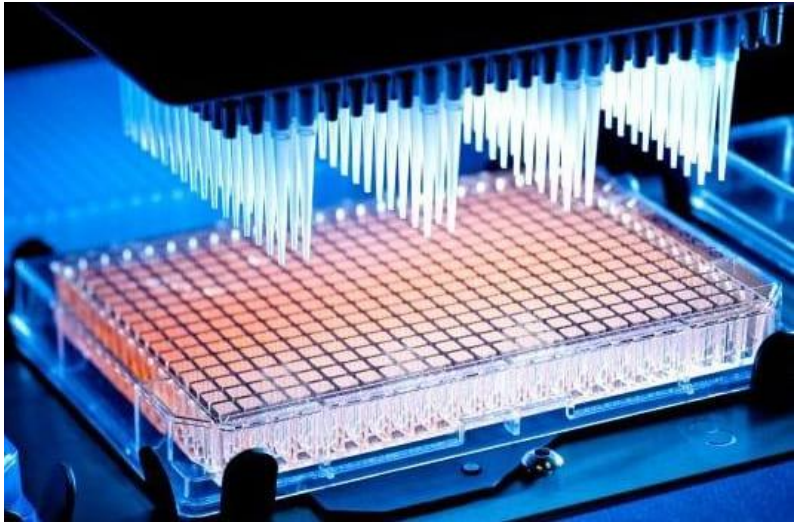
Off-target effect

- siRNAs can bind to off-target RNAs with mismatches
- Results in translational suppression or decay of the off-target RNAs
- The off-targeting of siRNAs can be minimized by formulating siRNA sequences with computational algorithms

Computational design

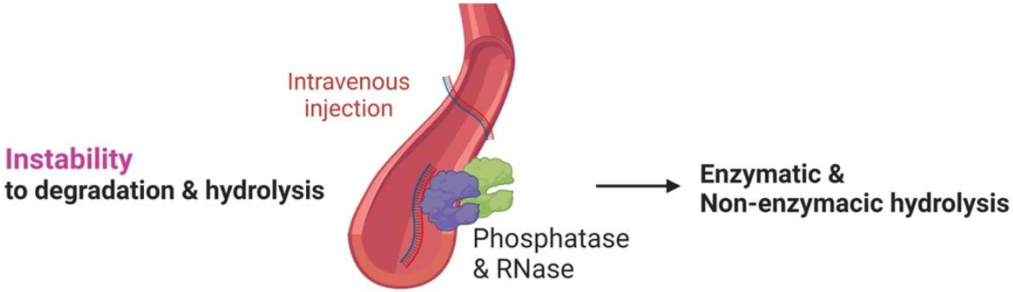
- in silico prediction of best siRNA candidates
- However, the predictive power is limited
  - Masking effects on the siRNA targeting sites due to the presence of secondary or tertiary structures or RNA-binding proteins, essentially limiting the access to the siRNA targeting site
  - For some target, there is a lack of correlation between the ability of RISC compatibility and its activity

- To compensate, extensive experimental screening of a large panel of siRNA candidates is needed



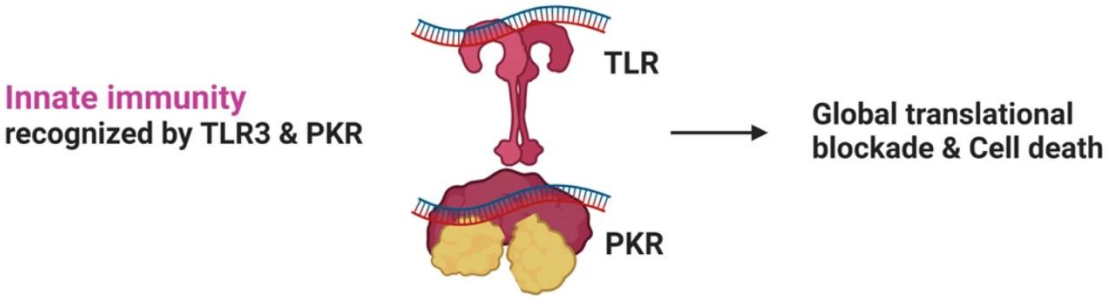
Instability

- RNAs are unstable and easily degraded by enzymatic and nonenzymatic hydrolysis
- Unmodified naked siRNAs delivered into organisms through intravenous injection can be rapidly degraded
- Degradation due to
  - 2′-hydroxyl (2′-OH)-mediated hydrolysis
  - Endonucleases
  - 5′ and 3′ exonucleases



Innate immune response

- dsRNA is immunogenic
- A high concentration of unformulated and unmodified siRNAs that can be recognized by Toll-like receptor 3 (TLR3) or serine/threonine protein kinase (PKR)
- This will activate innate immune responses, leading to global translational blockade and cell death

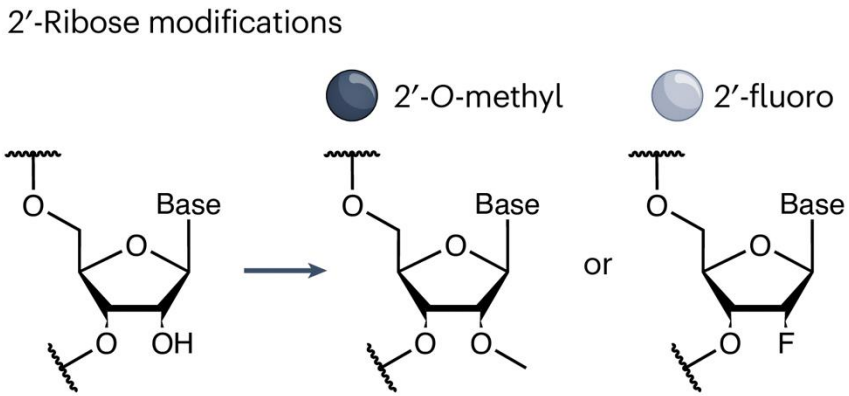


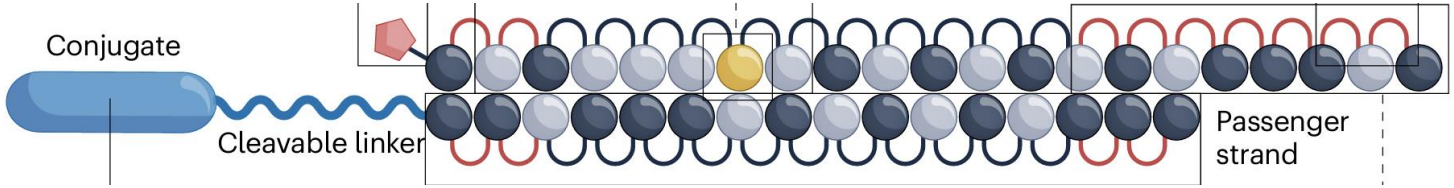
Chemical modifications

- Chemical stabilization
  - Enhance stability
  - Lower immune activation
  - Improve potency
- Unmodified siRNAs exhibit poor stability and are susceptible to immediate degradation
  - Over 50% within a minute in vivo
- Studies showed that a selectively small set of modifications are sufficient to generate nuclease resistance on the siRNA duplex and can also reduce immunogenicity

Scaffold stabilization – 2’ ribose

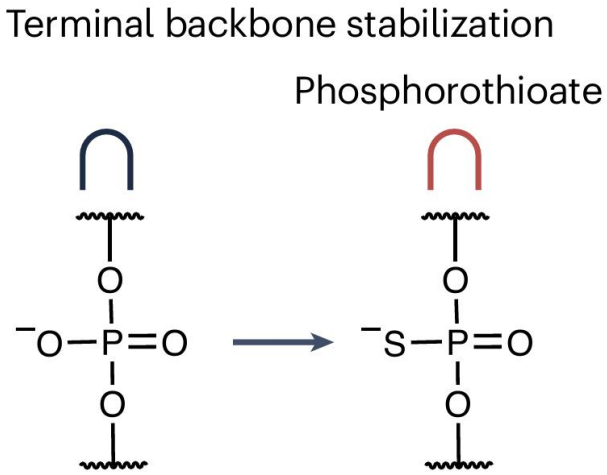
- 2’ hydroxyl group on RNA nucleotides make RNA vulnerable to 2′-hydroxyl (2′-OH)-mediated hydrolysis
- 2′-OMe and 2′-F can enhance serum stability and prolonged duration of action of siRNA

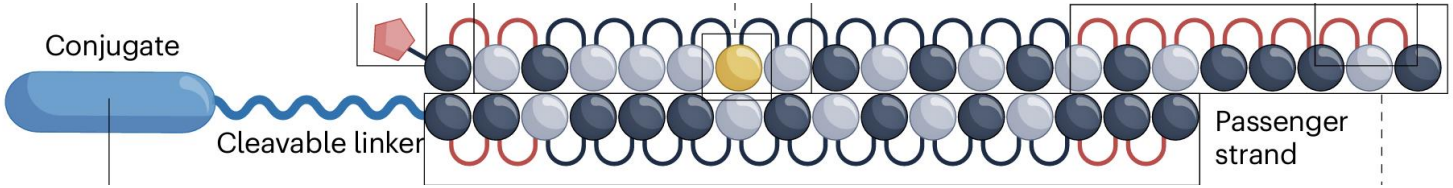




Scaffold stabilization - PS

- Phosphorothioates (PSs) can enhance siRNA stability in the body by making them resistant to degradation by nucleases



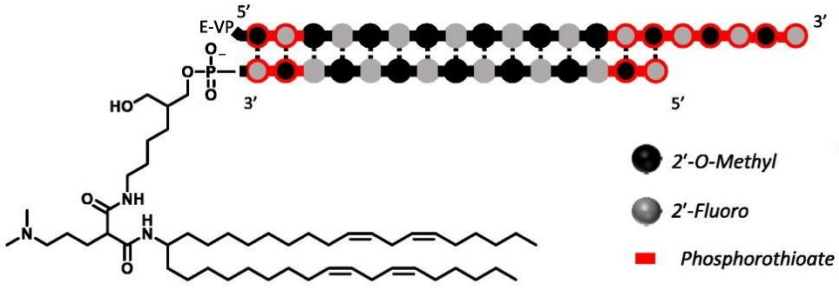


siRNA endosomal escape

- siRNA enters the cells primarily through endocytosis
- Like LNP-mRNA, siRNA is trapped in endosomes
- Endosomal escape is the rate-limiting step

- Slow endosome escape
  - An intracellular drug depot that gradually releases siRNA into the cytoplasm
  - This is the basis for the multi-month durability

- Fast endosome escape
  - Advantageous for siRNA therapies that require a rapid onset of silencing
  - Ionizable lipid conjugated to siRNA to enhance endosomal escape



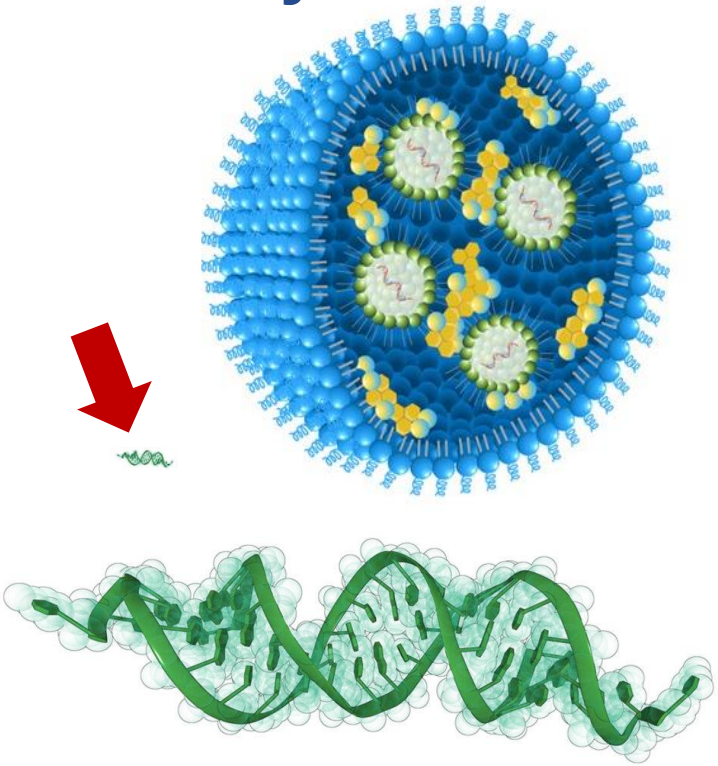
Ionizable DLin-MC3-DMA conjugated siRNAs

siRNA targeting strategy

What is the purpose of targeted delivery?

- siRNA is a much smaller molecule compared to mRNA
  - 7.5 nm in length and 2nm in diameter
- siRNA biodistribution is completed in a few hours after injection
- siRNA can get quickly cleared by the kidney

- Active targeting can
  - Reduce siRNA accumulation in unintended tissues
  - Decrease undesired side effects and toxicity
  - Reduce the dosage needed to achieve a potent gene silencing effect



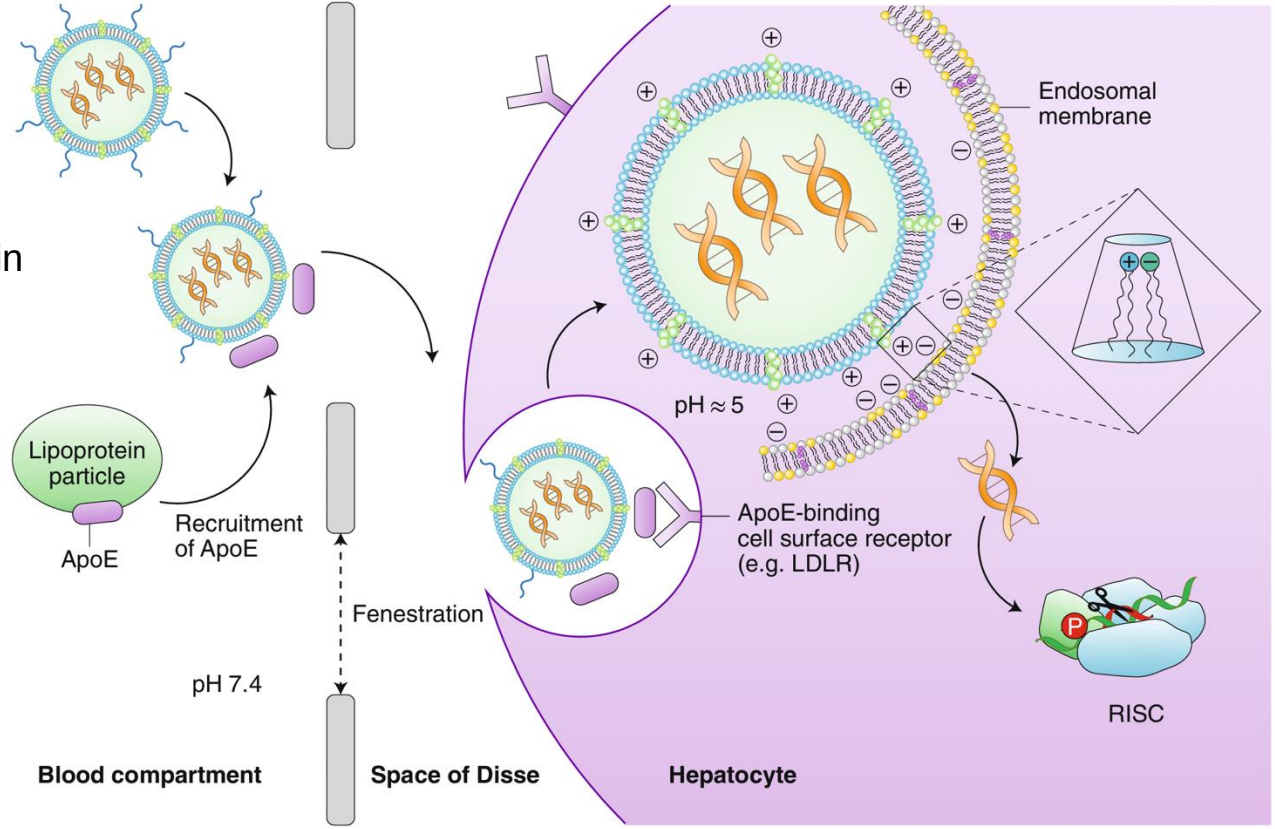
Targeting strategies

- Clinical Approved
  - Lipid nanoparticles
  - GalNAc conjugation

- Preclinical
  - Antibody drug conjugate
  - Lipid conjugate

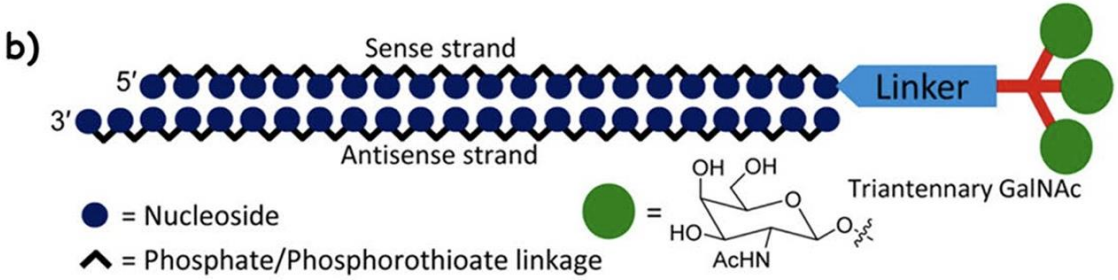
ONPATTRO delivery - LNP

- Recruitment of endogenous ApoE to the LNP surface
- Binding to low density lipoprotein receptors on hepatocytes
- Internalization of LNPs via endocytosis
- Protonation of the ionizable lipid in the low pH environment in the endosome
- Interaction of the positively charged ionizable lipid with negatively charged endogenous lipids
- Destabilization of the endosomal membrane
- Release of siRNA into the cytoplasm
- Intravenous administration



GalNAc conjugation

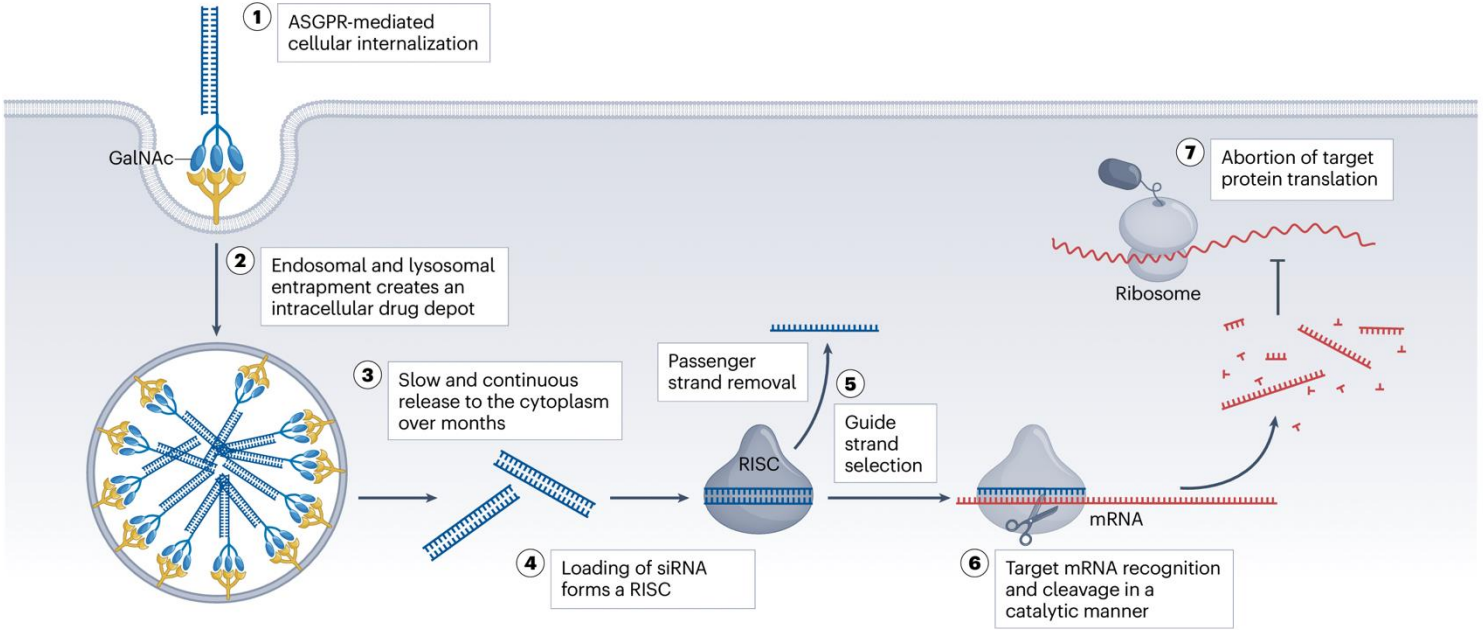
- N-acetylgalactosamine (GalNAc) is a sugar molecule
- 5 out of the 6 clinically approved siRNA drugs have GalNAc conjugation
- GalNAc is a ligand for the asialoglycoprotein receptor (ASGPR)
- ASGPR is exclusively and highly expressed on hepatocytes, allowing for liver-specific delivery
- Subcutaneous administration



siRNA–GalNAc Conjugate

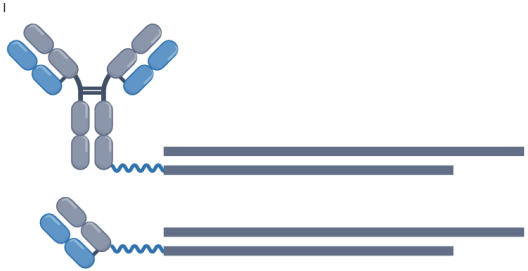
GalNAc-siRNA durability

- Slow release of GaLNAc-siRNA



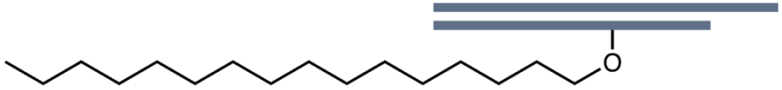
Targeting strategies under testing

- Antibody siRNA conjugate
  - Transferrin receptor 1 (TfR1) for skeletal, cardiac and smooth muscle delivery



Ab–Ab fragment conjugate

- Lipid siRNA conjugate
  - C16 conjugation
  - Robust clinical results upon local administration into the CNS, eye and lung

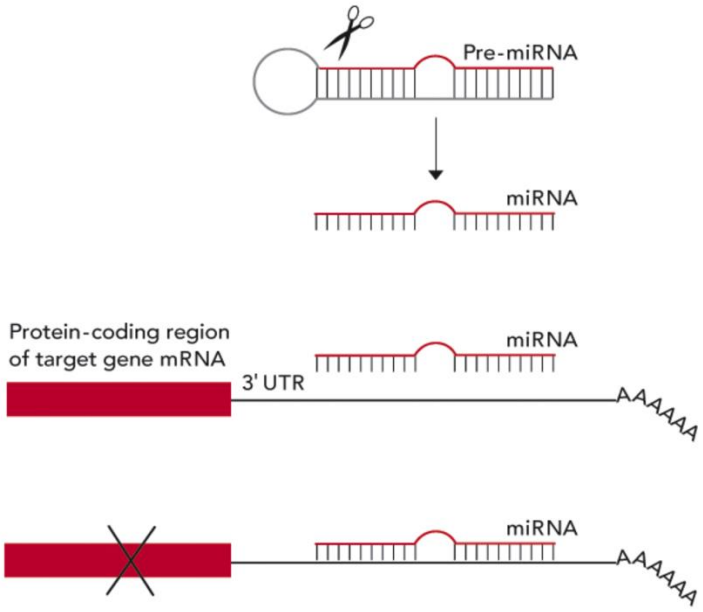


Lipid: 2′-*O*-hexadecyl

microRNA

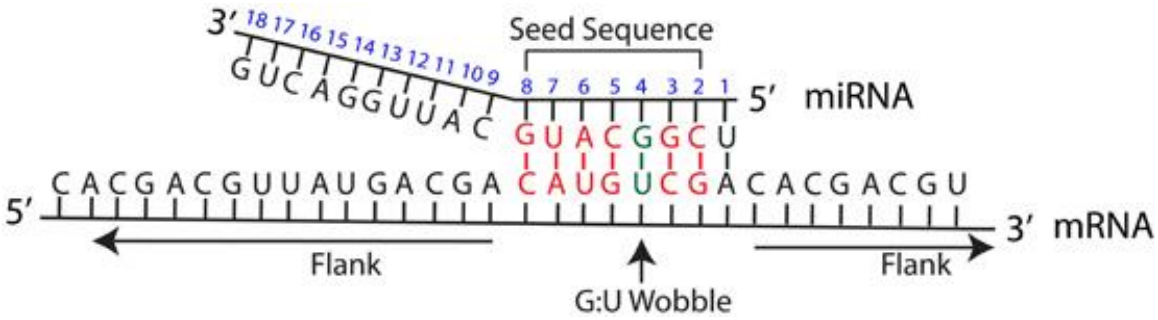
miRNA

- microRNA, miRNA, or miR
- miRNA are small non-coding RNAs and 18–24 nucleotides long
- miRNA can negatively regulate gene expression either by inhibiting mRNA translation or by promoting deadenylation and RNA degradation
- About 2500 miRNA genes have been identified in the human genome



miRNA

- miRNA binding regions are typically located in the 3′ UTR of mRNA
- miRNA typically does not bind perfectly to their target mRNA
- Seed sequence/region is at positions 2-7 from the miRNA 5’ end
    - Seed region needs to be perfectly complementary to the target mRNA



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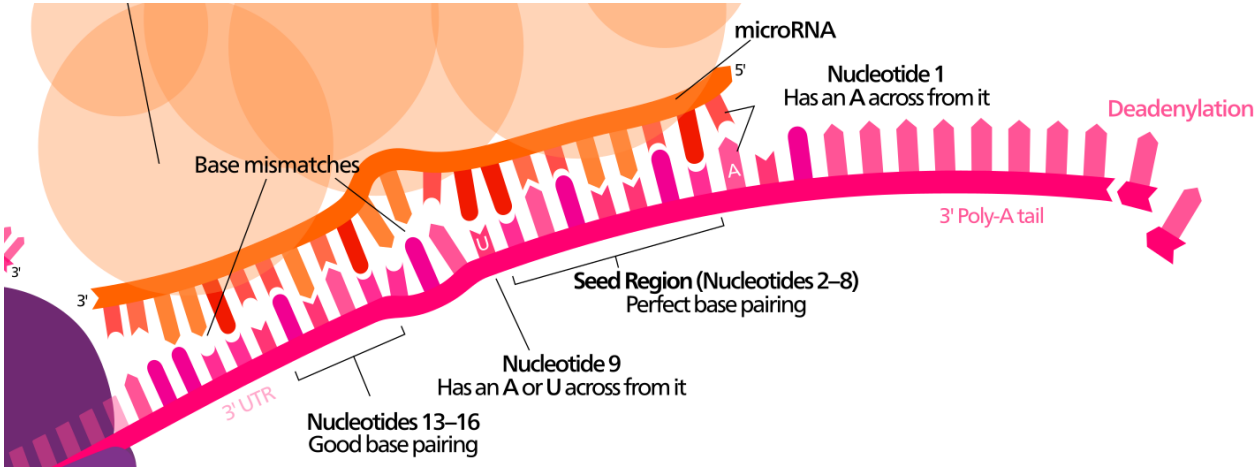
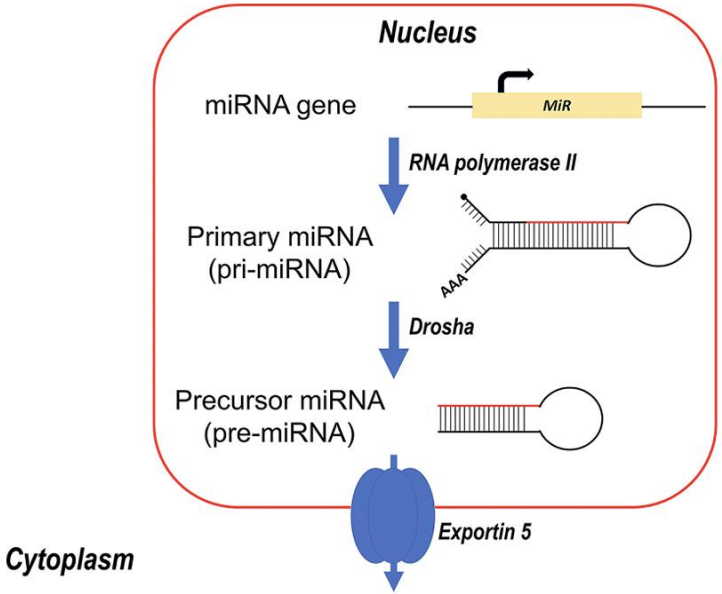
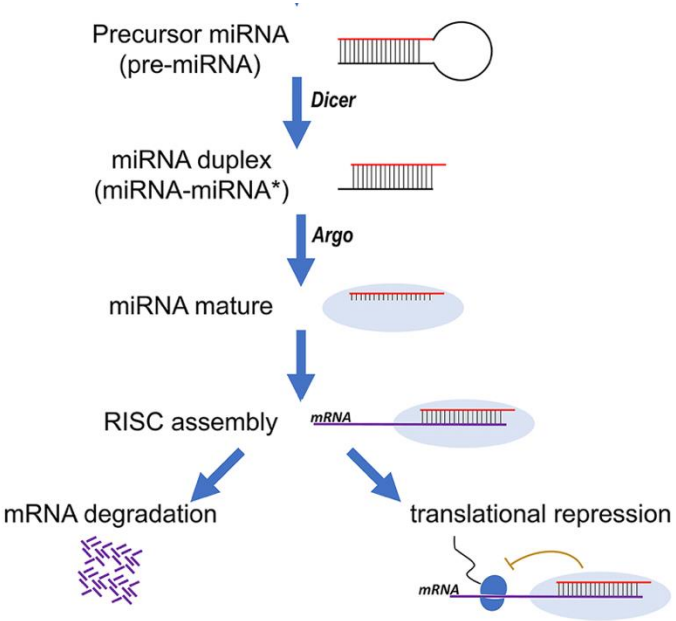


miRNA biogenesis

- miRNA genes are transcribed by RNA polymerase II into pri-miRNAs
- Drosha further process pri-miRNAs to pre-mRNAs
- pre-mRNAs are exported to the cytoplasm and processed into miRNA duplex by Dicer

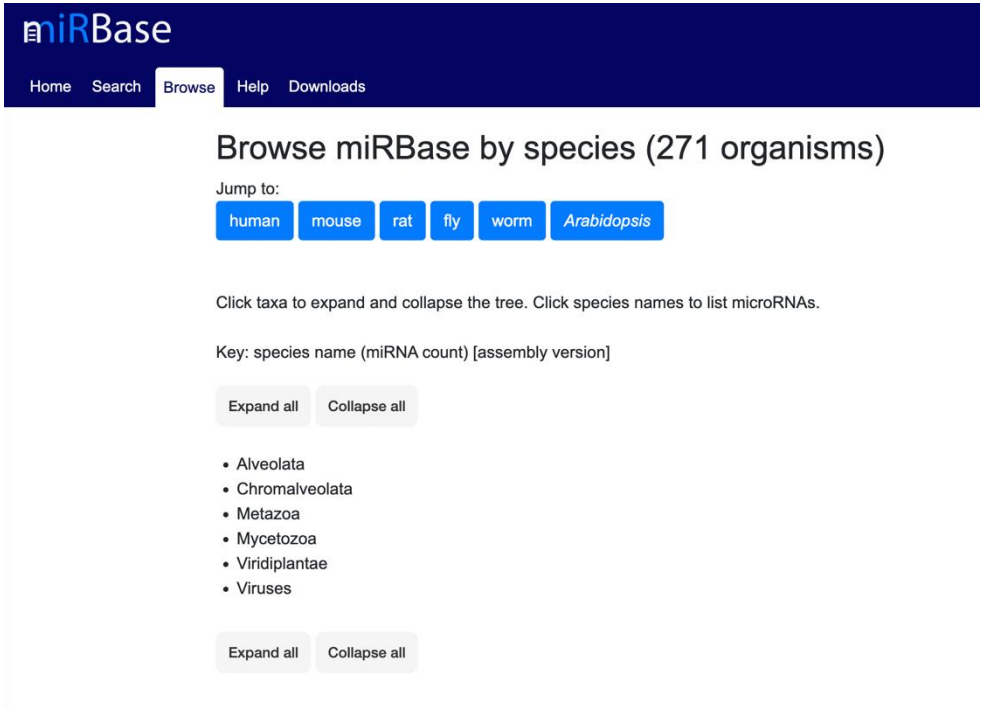


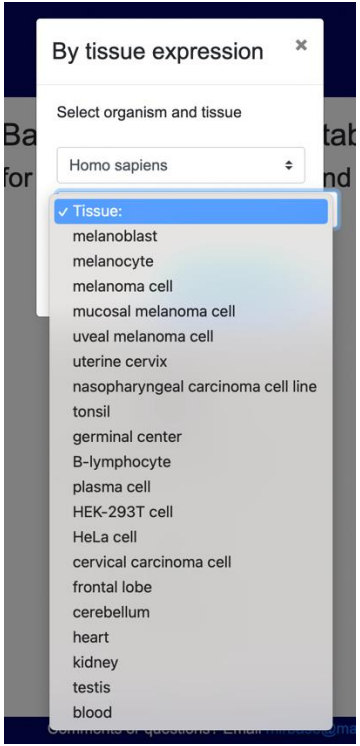


miRNA vs siRNA

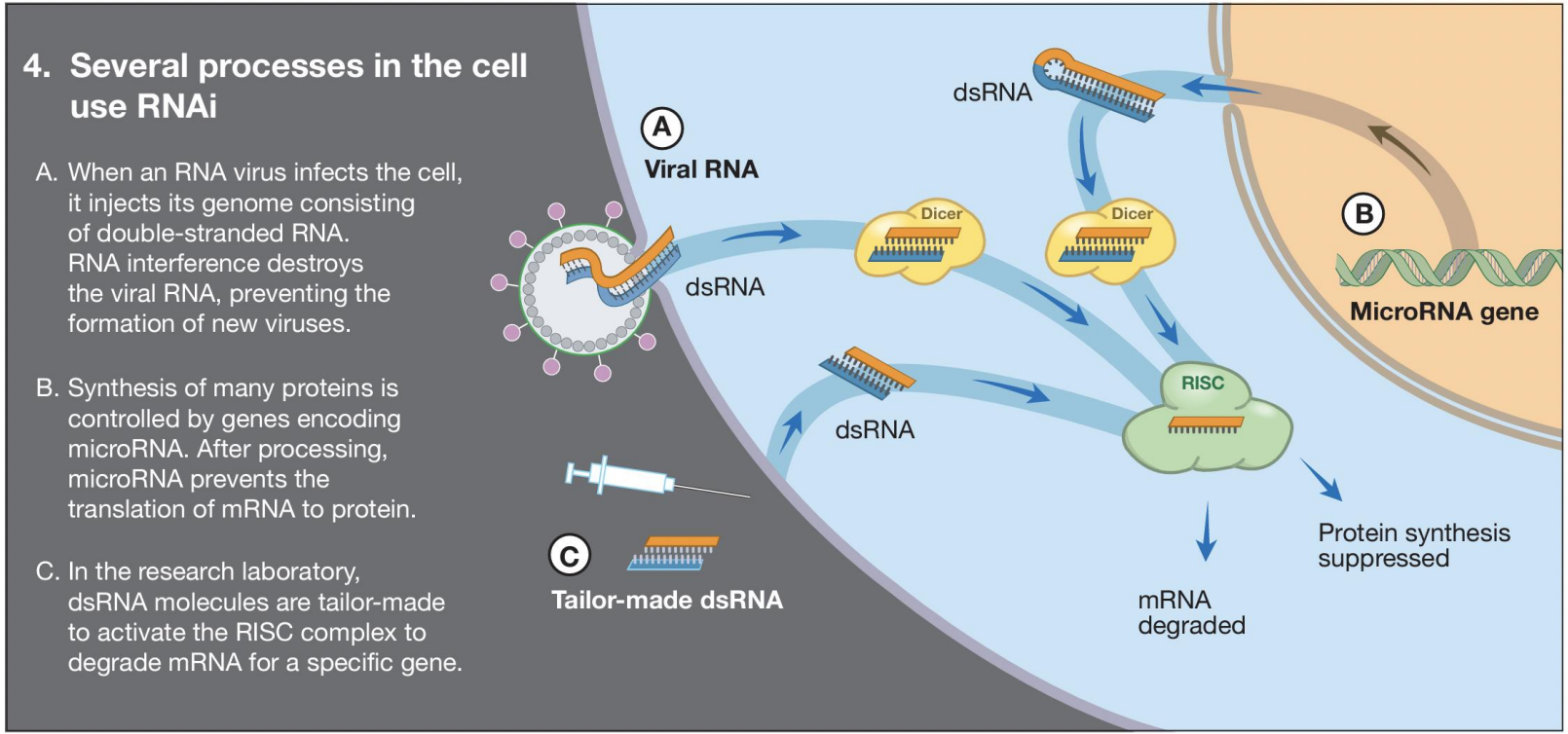
	miRNA	siRNA
Origin	Endogenous	Exogenous
Complementary	Partially complementary	Fully complementary
mRNA target	multiple	one
Mechanism of gene regulation	Translation inhibition mRNA deadenylation and degradation	Cleavage of mRNA

miRNA profiles
in different species and tissues





RNA interference (RNAi)



Discovery RNA interference (RNAi)

2006 Nobel Prize in Physiology or Medicine
Andrew Fire and Craig Mello

Photo: L. Cicero
Andrew Z. Fire

Photo: J. Mottern
Craig C. Mello

Discovery of miRNA

2024 Nobel Prize in Physiology or Medicine
Victor Ambros and Gary Ruvkun

Problem set #2

- **Due Friday, Jan 10th, 10PM EST**

Attendance